

Economics 1st-sem Notes

(Saurav Kumar)

Contents

1	Law of Demand	5
1.1	Demand Function	5
1.2	Basic Demand Curve	5
1.3	Movement Along vs. Shift of Demand	6
1.4	Example	6
2	Law of Supply	6
2.1	Supply Function	6
2.2	Basic Supply Curve	7
2.3	Movement Along vs. Shift of Supply	7
2.4	Why Supply Increases with Price?	7
3	Elasticity	7
3.1	General Concept	7
3.2	Price Elasticity of Demand (PED)	8
3.3	Types of Price Elasticity of Demand	8
3.4	Elasticity and Linear Demand	8
4	Market Equilibrium	9
4.1	Definition	9
4.2	Equilibrium Price and Quantity	9
4.3	Equilibrium Diagram	9
4.4	Shift in Demand and New Equilibrium	10
4.5	Shift in Supply and New Equilibrium	10
5	Government Intervention: Taxes, Subsidies, Price Controls	10
5.1	Specific Tax (Per-Unit Tax)	10
5.2	Subsidy	11
5.3	Price Ceiling and Price Floor	12
6	When the Market is Out of Balance	12
6.1	Surplus (Excess Supply)	12
6.2	Shortage (Excess Demand)	12
6.3	Diagram of Surplus and Shortage	13

7	Summary Tables	13
7.1	Determinants of Demand	13
7.2	Determinants of Supply	14
7.3	Types of Price Elasticity of Demand	14
7.4	Demand vs. Quantity Demanded	14
7.5	Supply vs. Quantity Supplied	15
7.6	Surplus vs. Shortage	15
7.7	Equilibrium Formula Summary	15
8	Government Price Controls	16
8.1	Price Ceiling (Maximum Price)	16
8.2	Price Floor (Minimum Price)	16
9	Consumer and Producer Surplus	17
9.1	Consumer Surplus (CS)	17
9.2	Producer Surplus (PS)	17
9.3	Social Surplus	17
10	Marginal Utility	18
10.1	Definition	18
10.2	Law of Diminishing Marginal Utility	18
10.3	Total Utility Curve	19
11	Total Utility, Marginal Utility and the Law of Diminishing Marginal Utility	19
11.1	Total Utility (TU)	19
11.2	Marginal Utility (MU)	19
11.3	Law of Diminishing Marginal Utility (DMU)	20
11.4	Example of DMU	20
11.5	Marginal Utility Curve	20
11.6	Total Utility Curve	20
12	Optimization and Utility Maximization	21
12.1	Unconstrained Optimization of a Single-Variable Function	21
12.2	Unconstrained Optimization of a Two-Variable Utility Function	22
12.3	Constrained Optimization: Utility Maximization	23
13	Consumer's Equilibrium	25
13.1	Deriving Equilibrium Using Total Differentiation	25
13.2	Total Differentiation of Budget Line	26
13.3	Consumer Equilibrium Condition	26
13.4	Equilibrium with Many Goods	26
13.5	Graphical Representation: IC and Budget Line	27
14	Supply Curve: Theory	27

15 Perfect Competition	28
16 Worked Numerical Examples	28
16.1 Example 1: Consumer Equilibrium Using MU per Rupee	28
16.2 Example 2: Utility Maximization with Lagrangian	29
16.3 Example 3: Firm Supply from Profit Maximization	29
17 Lagrangian Diagram for Consumer Choice	30
18 Supply Curve with MC and ATC	30
18.1 Firm's Short-Run Supply	30
19 Consumer Equilibrium: Summary Table	31
20 Revision Notes (Quick Summary)	31
21 Production Theory	32
21.1 Two-Input Production Function	32
21.2 Assumptions of the Production Function	33
21.3 Isoquant Diagram	33
21.4 Examples of Production Functions	33
22 Isoquants and MRTS	34
22.1 Properties of Isoquants	34
22.2 Isoquant Diagram	34
22.3 Marginal Rate of Technical Substitution (MRTS)	34
22.4 Law of Diminishing MRTS	35
22.5 Example	35
23 Cost Minimisation and $MRTS = w/r$	35
23.1 Isocost Line	36
23.2 Condition for Cost Minimisation	36
23.3 Cost Minimisation Diagram	36
23.4 Numerical Example	37
24 Cobb–Douglas Production Function and Returns to Scale	37
24.1 Interpretation of Parameters	37
24.2 Returns to Scale	38
24.3 Technology Shifts and Isoquants	38
24.4 Example: Effect of Technology	38
25 Perfect Substitutes and Perfect Complements	39
25.1 Perfect Substitutes	39
25.2 Perfect Complements (Leontief)	39

26 Short-Run and Long-Run Costs	40
26.1 Short-Run Costs	40
26.2 Average and Marginal Costs	41
26.3 Short-Run Cost Curve Diagram	41
26.4 TFC, TVC, and TC Curves	42
26.5 Long-Run Costs	42
26.6 LRAC and SRAC Diagram	42
26.7 Economies and Diseconomies of Scale	43
27 Perfect Competition	43
27.1 Price Taking and Revenue	43
27.2 Profit Maximisation: $P = MC$	44
27.3 Short-Run Equilibrium of the Firm	44
27.4 Shutdown Condition	44
27.5 Firm's Supply Curve	44
27.6 Long-Run Equilibrium	45
28 Profit Maximisation and Shutdown Decision	45
28.1 Profit Function	45
28.2 First-Order Condition (FOC)	46
28.3 Second-Order Condition (SOC)	46
28.4 Profit Maximisation Diagram	47
28.5 Total Revenue, Total Cost, and Profit	47
28.6 Shutdown Decision	47
28.7 Summary of Cases	48
29 Industry–Firm Equilibrium, Profit, Loss and Long-Run Adjustment	48
29.1 Step 1: Industry Equilibrium and Firm Price	48
29.2 Step 2: Adding ATC and AVC to Determine Profit or Loss	49
29.3 Step 4: Total Revenue, Total Cost and Profit	50
29.4 Step 6–9: Demand Decreases and the Firm Makes a Loss	50
29.5 Shutdown Rule (Extreme Negative Case)	50
29.6 Long-Run Equilibrium of a Competitive Industry	51

1 Law of Demand

Definition (ceteris paribus): The higher the price of a commodity, the lower will be the quantity demanded, and vice versa.

1.1 Demand Function

General form:

$$Q_d = f(P), \quad \frac{dQ_d}{dP} < 0$$

Simple linear demand:

$$Q_d = a - bP, \quad a > 0, b > 0$$

1.2 Basic Demand Curve

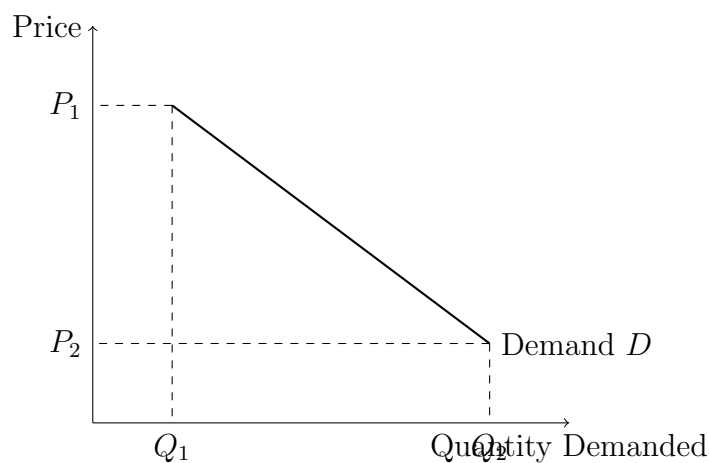
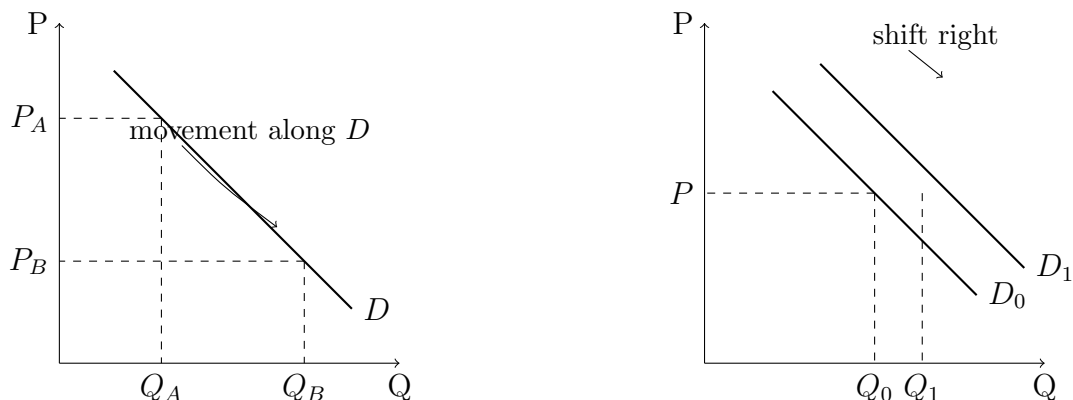


Figure 1: Downward-sloping demand curve

1.3 Movement Along vs. Shift of Demand



(a) Movement along demand (price change)

(b) Shift of demand (non-price factor)

Figure 2: Demand: movement vs. shift

1.4 Example

Suppose

$$Q_d = 100 - 2P.$$

If P rises from 20 to 25:

$$Q_{d1} = 100 - 2(20) = 60, \quad Q_{d2} = 100 - 2(25) = 50.$$

Quantity demanded falls from 60 units to 50 units.

2 Law of Supply

Definition (*ceteris paribus*): The higher the price of a commodity, the greater will be the quantity supplied.

2.1 Supply Function

$$Q_s = g(P), \quad \frac{dQ_s}{dP} > 0$$

Linear form:

$$Q_s = c + dP, \quad c \geq 0, \quad d > 0$$

2.2 Basic Supply Curve

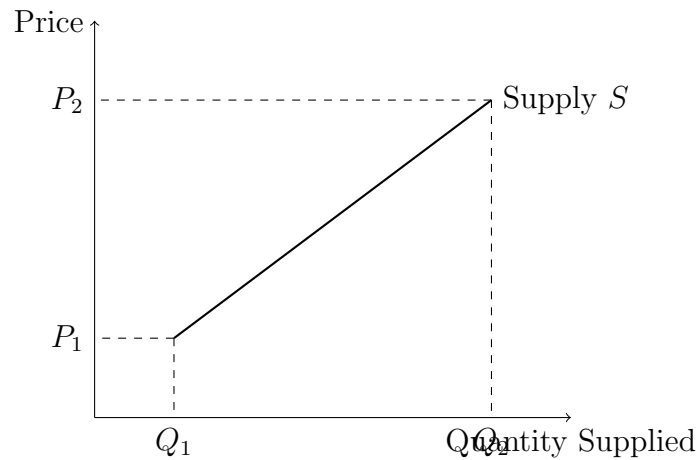


Figure 3: Upward-sloping supply curve

2.3 Movement Along vs. Shift of Supply

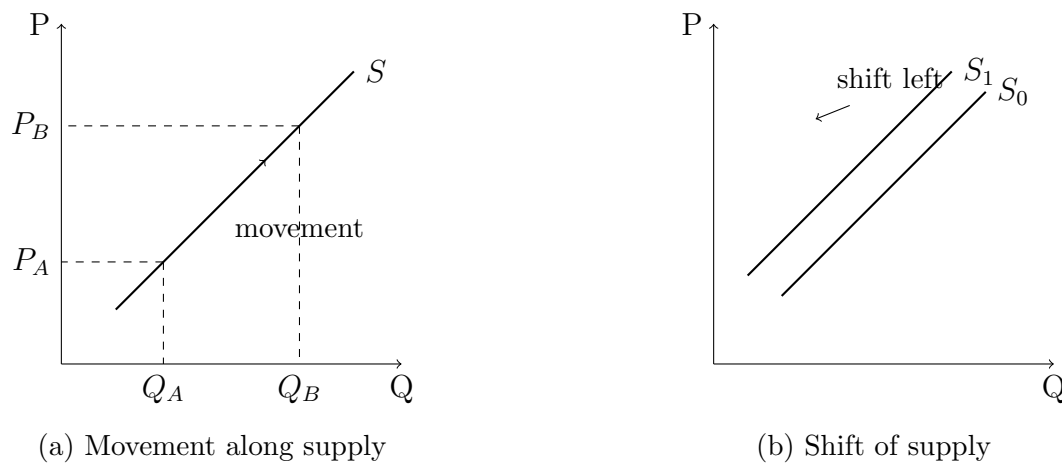


Figure 4: Supply: movement vs. shift

2.4 Why Supply Increases with Price?

- **Profit motive:** higher price $\rightarrow$ higher potential profit $\rightarrow$ producers supply more.
- **New entrants:** higher price attracts new firms into the industry.

3 Elasticity

3.1 General Concept

Elasticity measures the responsiveness of one variable to changes in another variable.

3.2 Price Elasticity of Demand (PED)

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{\Delta Q_d}{Q_d} \bigg/ \frac{\Delta P}{P}$$

Point Elasticity (using calculus)

If $Q_d = f(P)$ is differentiable:

$$E_d = \frac{dQ_d}{dP} \cdot \frac{P}{Q_d}$$

Arc Elasticity (between two points)

Between (P_1, Q_1) and (P_2, Q_2) :

$$E_d = \frac{\frac{Q_2 - Q_1}{Q_1 + Q_2}}{\frac{P_2 - P_1}{\frac{P_1 + P_2}{2}}}$$

3.3 Types of Price Elasticity of Demand

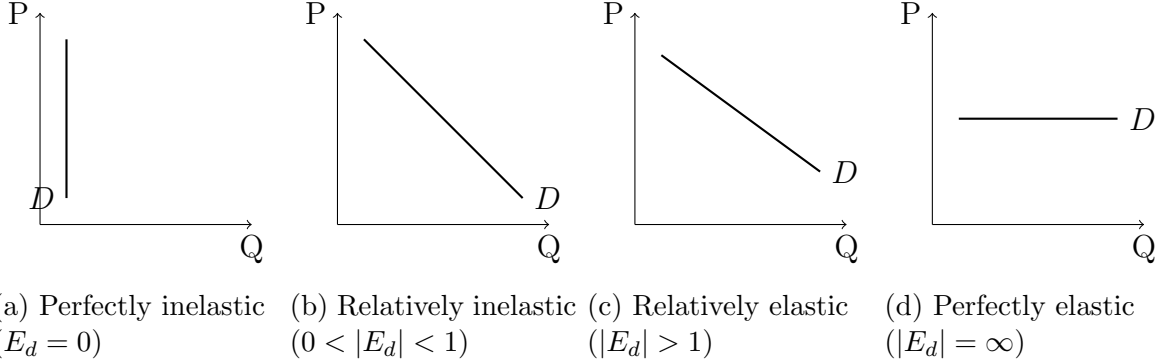


Figure 5: Different demand elasticities

3.4 Elasticity and Linear Demand

For a straight-line demand $Q_d = a - bP$:

$$E_d = \frac{dQ_d}{dP} \cdot \frac{P}{Q_d} = (-b) \cdot \frac{P}{a - bP}$$

Elasticity varies at different points on the same straight line.

4 Market Equilibrium

4.1 Definition

Market equilibrium is a situation where quantity demanded equals quantity supplied at a particular price.

$$Q_d = Q_s$$

With linear functions:

$$Q_d = a - bP, \quad Q_s = c + dP.$$

4.2 Equilibrium Price and Quantity

Set $Q_d = Q_s$:

$$a - bP = c + dP \Rightarrow a - c = (b + d)P.$$

$$P^* = \frac{a - c}{b + d}$$

Substituting into either function:

$$Q^* = a - bP^* = c + dP^* = \frac{ad + bc}{b + d}$$

4.3 Equilibrium Diagram

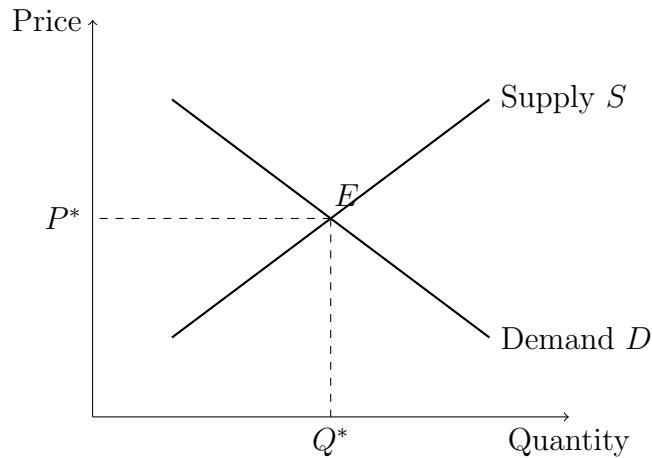


Figure 6: Market equilibrium

4.4 Shift in Demand and New Equilibrium

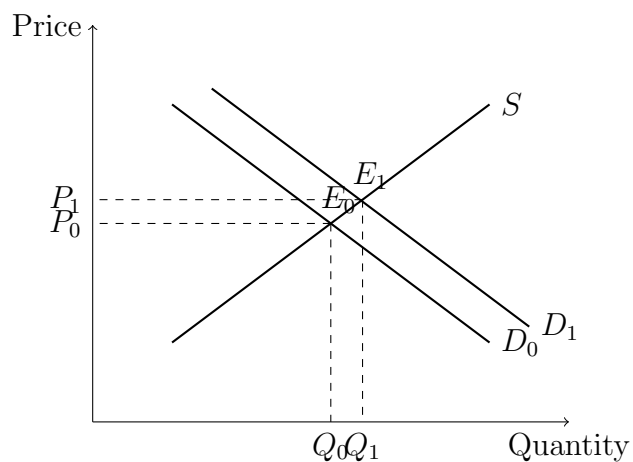


Figure 7: Increase in demand: higher P and higher Q

4.5 Shift in Supply and New Equilibrium

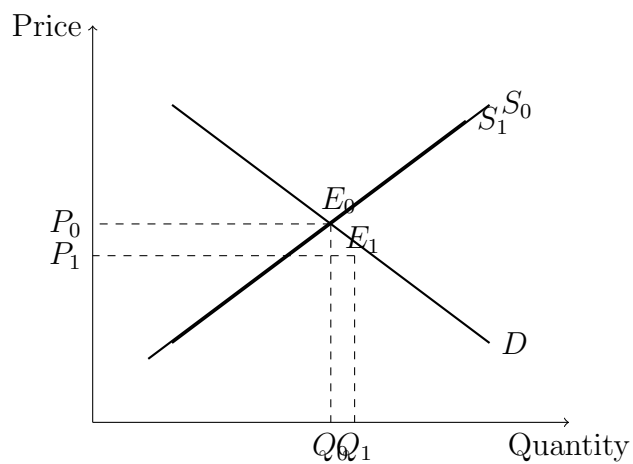


Figure 8: Increase in supply: lower P , higher Q

5 Government Intervention: Taxes, Subsidies, Price Controls

5.1 Specific Tax (Per-Unit Tax)

A per-unit tax shifts the supply curve upward by the amount of the tax t .

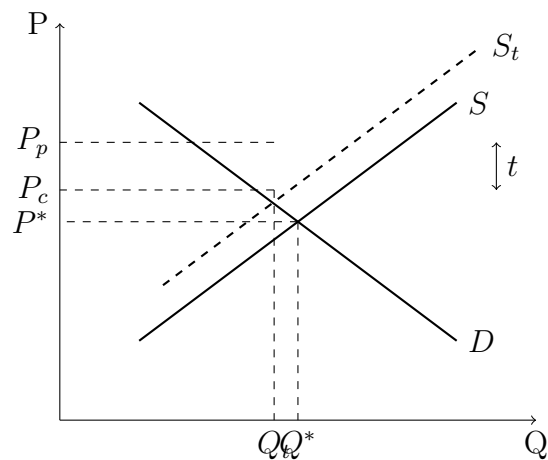


Figure 9: Per-unit tax: $S \rightarrow S_t$

5.2 Subsidy

A per-unit subsidy shifts the supply curve downward by amount s (producers receive more than the market price).

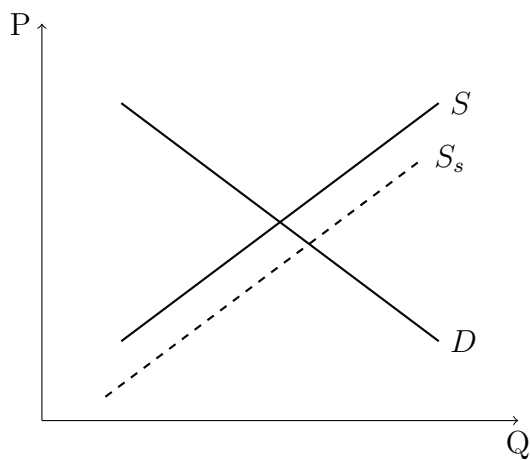
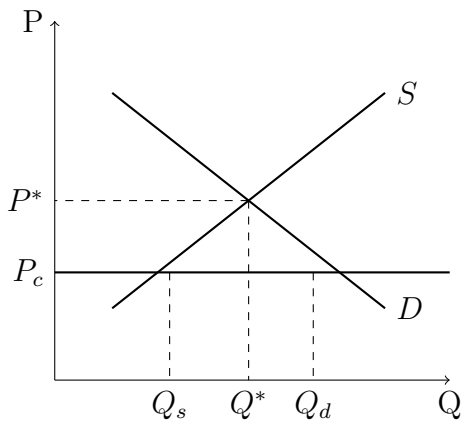
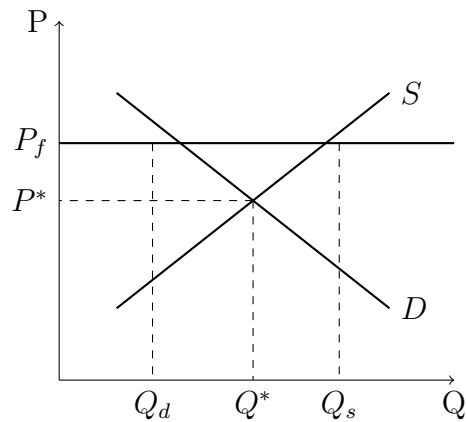


Figure 10: Subsidy: supply shifts downward

5.3 Price Ceiling and Price Floor



(a) Price ceiling ($P_c < P^*$): shortage



(b) Price floor ($P_f > P^*$): surplus

Figure 11: Price controls

6 When the Market is Out of Balance

6.1 Surplus (Excess Supply)

A **surplus** occurs when the market price is above the equilibrium price:

$$P > P^* \Rightarrow Q_s > Q_d.$$

Magnitude of surplus:

$$\text{Surplus} = Q_s - Q_d.$$

6.2 Shortage (Excess Demand)

A **shortage** occurs when the market price is below the equilibrium price:

$$P < P^* \Rightarrow Q_d > Q_s.$$

Magnitude of shortage:

$$\text{Shortage} = Q_d - Q_s.$$

6.3 Diagram of Surplus and Shortage

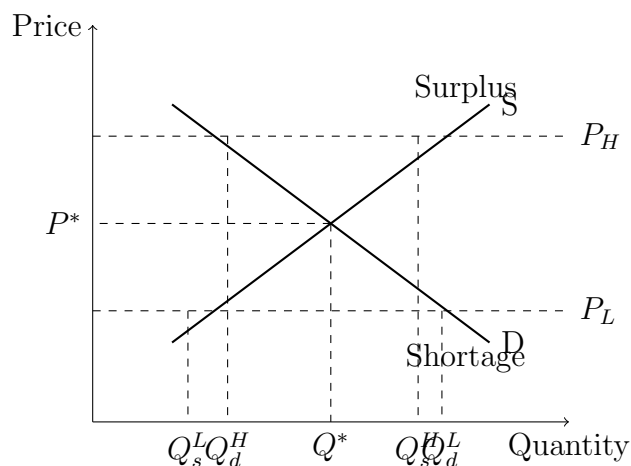


Figure 12: Surplus and shortage at non-equilibrium prices

7 Summary Tables

7.1 Determinants of Demand

Factor	Effect on Demand
Price of the good	Movement along demand curve
Income	Normal good: income $\uparrow \Rightarrow D \uparrow$ Inferior good: income $\uparrow \Rightarrow D \downarrow$
Price of substitutes	Price of substitute $\uparrow \Rightarrow D \uparrow$
Price of complements	Price of complement $\uparrow \Rightarrow D \downarrow$
Tastes/preferences	More preference $\Rightarrow D \uparrow$
Expectations	Higher expected future price $\Rightarrow D \uparrow$ now

Table 1: Determinants of demand

7.2 Determinants of Supply

Factor	Effect on Supply
Own price of the good	Movement along supply curve
Input prices	Input price $\uparrow \Rightarrow S \downarrow$
Technology	Better technology $\Rightarrow S \uparrow$
Number of sellers	More firms $\Rightarrow S \uparrow$
Taxes	Specific tax $\Rightarrow S \downarrow$ (shift up)
Subsidies	Subsidy $\Rightarrow S \uparrow$ (shift down)
Expectations	Higher expected future price may reduce current S

Table 2: Determinants of supply

7.3 Types of Price Elasticity of Demand

Type	Numerical Range	Description
Perfectly inelastic	$E_d = 0$	Quantity does not respond to price
Relatively inelastic	$0 < E_d < 1$	Quantity responds less than proportionately
Unitary elastic	$ E_d = 1$	Equal percentage change in Q and P
Relatively elastic	$ E_d > 1$	Quantity responds more than proportionately
Perfectly elastic	$ E_d = \infty$	Any small price change causes infinite Q change

Table 3: Types of price elasticity of demand

7.4 Demand vs. Quantity Demanded

	Demand	Quantity Demanded
Meaning	Entire relationship between P and Q_d	Specific amount at a given price
Graph	Whole demand curve	Point on demand curve
Cause of change	Non-price determinants (income, tastes, etc.)	Change in price of the good itself
Graphical effect	Shift of demand curve	Movement along curve

Table 4: Difference between demand and quantity demanded

7.5 Supply vs. Quantity Supplied

	Supply	Quantity Supplied
Meaning	Entire relationship between P and Q_s	Amount supplied at a particular price
Graph	Whole supply curve	Point on supply curve
Cause of change	Non-price determinants (technology, input prices, etc.)	Change in price of the good
Graphical effect	Shift of supply curve	Movement along curve

Table 5: Difference between supply and quantity supplied

7.6 Surplus vs. Shortage

	Surplus	Shortage
Condition	$P > P^*$	$P < P^*$
Quantity relation	$Q_s > Q_d$	$Q_d > Q_s$
Market pressure	Price tends to fall	Price tends to rise
Formula	$Q_s - Q_d$	$Q_d - Q_s$

Table 6: Surplus and shortage

7.7 Equilibrium Formula Summary

Concept	Formula
Demand function	$Q_d = a - bP$
Supply function	$Q_s = c + dP$
Equilibrium price	$P^* = \frac{a - c}{b + d}$
Equilibrium quantity	$Q^* = \frac{ad + bc}{b + d}$
Point price elasticity of demand	$E_d = \frac{dQ_d}{dP} \cdot \frac{P}{Q_d}$
Arc elasticity of demand	$E_d = \frac{(Q_1 + Q_2)/2}{P_2 - P_1}$
Surplus	$Q_s - Q_d$ (when $P > P^*$)
Shortage	$Q_d - Q_s$ (when $P < P^*$)

Table 7: Key formulas for equilibrium and elasticity

8 Government Price Controls

8.1 Price Ceiling (Maximum Price)

A **price ceiling** is a government-imposed maximum price set below equilibrium to make goods affordable.

$$P_c < P^*$$

At P_c , quantity demanded exceeds quantity supplied:

$$Q_d > Q_s \Rightarrow \text{Shortage}$$

Example

Equilibrium price of wheat = 40/kg. Government sets a ceiling at 25/kg.

$$Q_d = 500, Q_s = 300$$

$$\text{Shortage} = 200$$

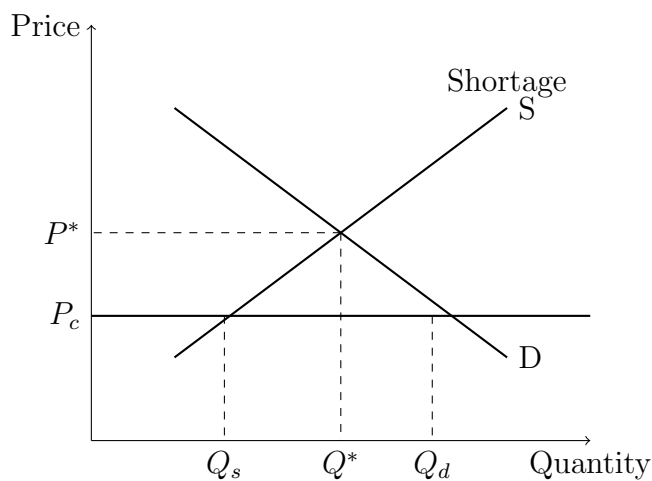


Figure 13: Price Ceiling

8.2 Price Floor (Minimum Price)

A **price floor** is a minimum price set by the government above equilibrium.

$$P_f > P^*$$

At P_f :

$$Q_s > Q_d \Rightarrow \text{Surplus}$$

Example

Equilibrium wheat price = 1600. MSP (floor price) = 2000.

$$Q_s = 60000, \quad Q_d = 40000$$

$$\text{Surplus} = 20000$$

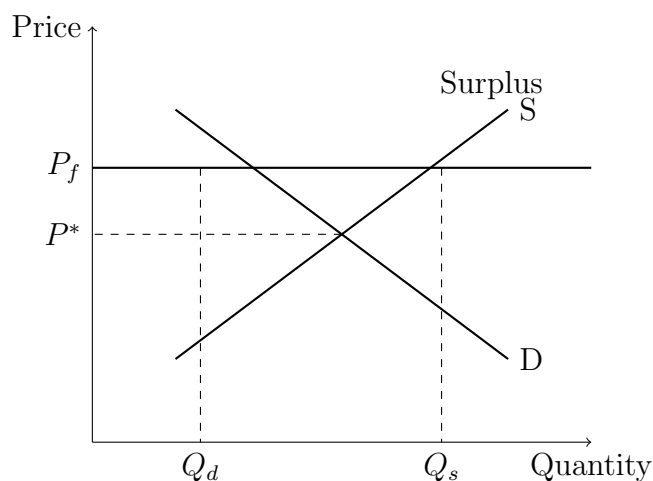


Figure 14: Price Floor

9 Consumer and Producer Surplus

9.1 Consumer Surplus (CS)

CS is the difference between the maximum price a consumer is willing to pay and the actual price.

$$CS = \frac{1}{2}(P_{\max} - P^*)Q^*$$

9.2 Producer Surplus (PS)

PS is the difference between the price actually received and the minimum price producers are willing to accept.

$$PS = \frac{1}{2}(P^* - P_{\min})Q^*$$

9.3 Social Surplus

$$\text{Social Surplus} = CS + PS$$

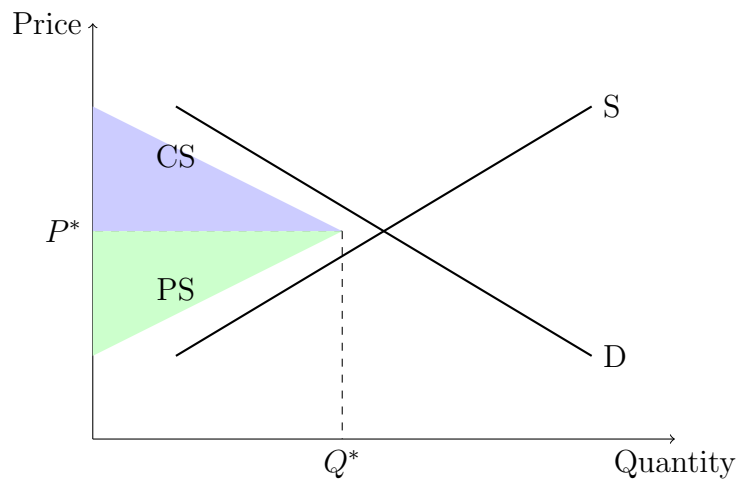


Figure 15: Consumer and Producer Surplus

10 Marginal Utility

10.1 Definition

Marginal utility (MU) is the extra satisfaction from consuming one more unit.

$$MU = \frac{\Delta TU}{\Delta Q}$$

10.2 Law of Diminishing Marginal Utility

$$MU \downarrow \text{ as } Q \uparrow$$

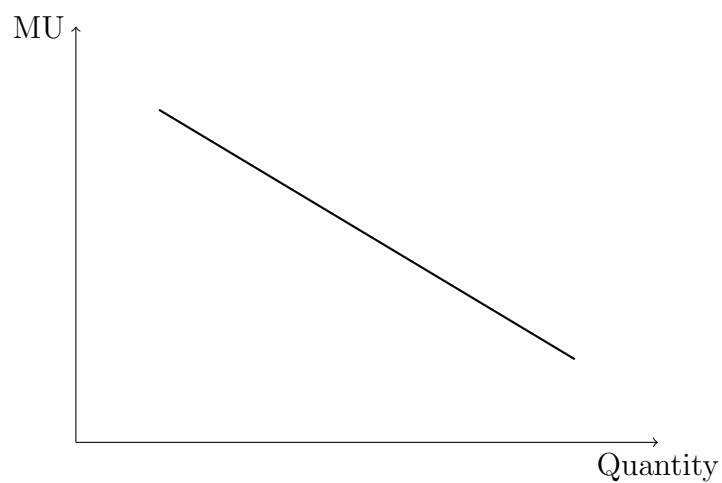


Figure 16: Marginal Utility Curve

10.3 Total Utility Curve

TU increases at a decreasing rate.

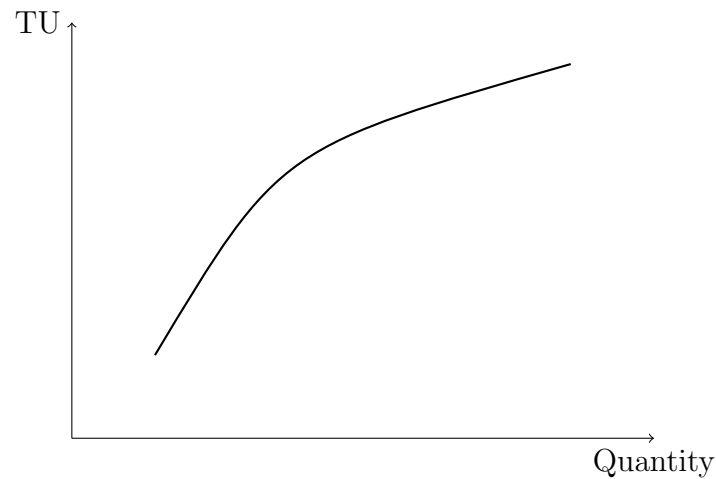


Figure 17: Total Utility Curve

11 Total Utility, Marginal Utility and the Law of Diminishing Marginal Utility

11.1 Total Utility (TU)

Total utility refers to the total satisfaction a consumer receives from consuming a certain number of units of a good.

- TU generally increases as more units are consumed.
- However, the increase in TU becomes smaller for each additional unit.

Example:

If a consumer eats pizza slices:

$$TU_1 = 20, \quad TU_2 = 35, \quad TU_3 = 47$$

11.2 Marginal Utility (MU)

Marginal Utility is the additional satisfaction obtained from consuming one more unit of a good.

$$MU = \frac{\Delta TU}{\Delta Q}$$

Example:

If $TU_2 = 35$ and $TU_1 = 20$, then

$$MU_2 = TU_2 - TU_1 = 15$$

11.3 Law of Diminishing Marginal Utility (DMU)

The law states:

As more units of a good are consumed, the marginal utility obtained from each additional unit keeps decreasing.

$$MU \downarrow \text{ as } Q \uparrow$$

This occurs due to saturation, limited human capacity, and reduced intensity of want.

11.4 Example of DMU

Consider the satisfaction a person gets from drinking glasses of juice:

$$MU_1 = 30, MU_2 = 25, MU_3 = 20, MU_4 = 15, MU_5 = 10$$

The marginal utility continually falls, illustrating DMU.

11.5 Marginal Utility Curve

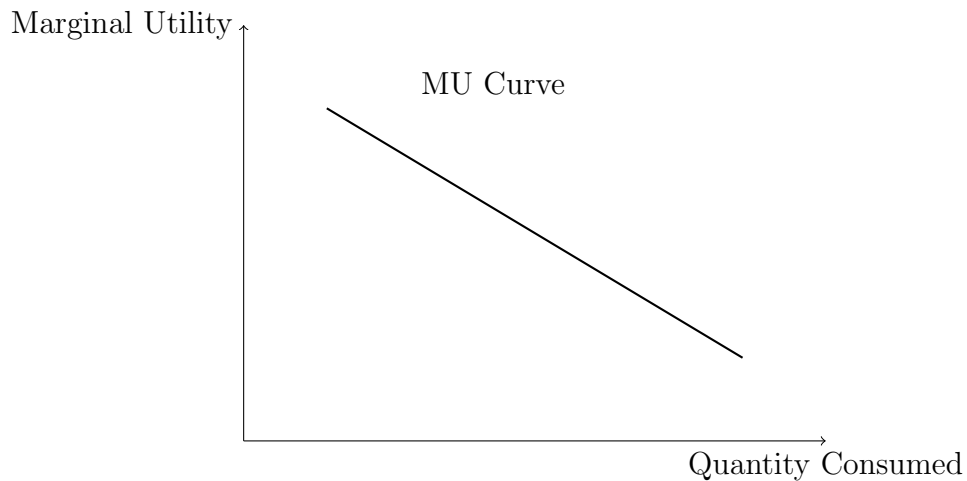


Figure 18: Marginal Utility Curve

11.6 Total Utility Curve

Total utility initially increases but at a decreasing rate. It reaches a maximum when $MU = 0$ and may fall afterward if consumption continues.

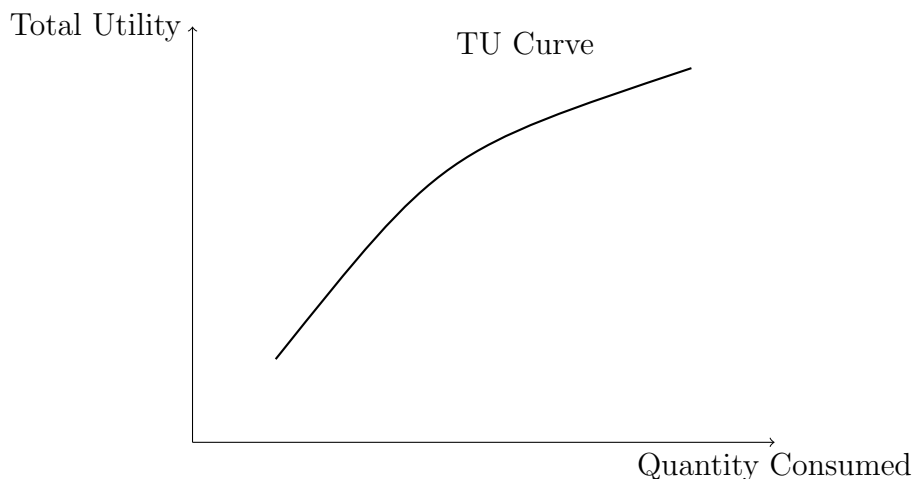


Figure 19: Total Utility Curve

12 Optimization and Utility Maximization

12.1 Unconstrained Optimization of a Single-Variable Function

Suppose we want to maximize a function

$$y = f(x).$$

First-Order Condition (FOC) – Necessary Condition

To find a maximum (or minimum), we differentiate with respect to x :

$$\frac{dy}{dx} = f'(x).$$

At an interior maximum or minimum, the slope must be zero:

$$f'(x) = 0.$$

If the solution is x^* , then

$$f'(x^*) = 0.$$

This gives us a *critical point* but does not tell us whether it is a maximum or a minimum.

Second-Order Condition (SOC) – Sufficient Condition for Maximum

We take the second derivative:

$$f''(x).$$

For a **maximum**, the curvature must be negative at x^* :

$$f''(x^*) < 0.$$

For a **minimum**, we require

$$f''(x^*) > 0.$$

Example

Let

$$f(x) = 10x - x^2.$$

FOC:

$$f'(x) = 10 - 2x = 0 \quad \Rightarrow \quad x^* = 5.$$

SOC:

$$f''(x) = -2 < 0 \quad \Rightarrow \quad x^* = 5 \text{ is a maximum.}$$

The maximum value of the function is

$$f(5) = 10(5) - 5^2 = 50 - 25 = 25.$$

12.2 Unconstrained Optimization of a Two-Variable Utility Function

Now consider a utility function of two goods:

$$U = U(x, y).$$

First-Order Conditions (FOC)

To find stationary points, we take partial derivatives:

$$\frac{\partial U}{\partial x} = U_x(x, y) = 0,$$

$$\frac{\partial U}{\partial y} = U_y(x, y) = 0.$$

Solving this system gives candidate points (x^*, y^*) for maxima or minima.

Second-Order Condition (SOC)

We compute the **Hessian matrix**:

$$H = \begin{bmatrix} U_{xx} & U_{xy} \\ U_{yx} & U_{yy} \end{bmatrix},$$

where

$$U_{xx} = \frac{\partial^2 U}{\partial x^2}, \quad U_{yy} = \frac{\partial^2 U}{\partial y^2}, \quad U_{xy} = \frac{\partial^2 U}{\partial x \partial y}, \quad U_{yx} = \frac{\partial^2 U}{\partial y \partial x}.$$

Young's Theorem If $U(x, y)$ has continuous second-order partial derivatives, then

$$U_{xy} = U_{yx}.$$

So the Hessian is symmetric.

Condition for a Maximum (Negative Definiteness) For (x^*, y^*) to be a **maximum**, the Hessian must be *negative definite*. In two dimensions this means:

$$U_{xx}(x^*, y^*) < 0, \quad U_{yy}(x^*, y^*) < 0,$$

and the determinant

$$|H| = U_{xx}U_{yy} - (U_{xy})^2 > 0.$$

These conditions ensure that $U(x, y)$ is locally concave at (x^*, y^*) , guaranteeing maximum utility.

12.3 Constrained Optimization: Utility Maximization

A consumer chooses x and y to maximize utility subject to a budget constraint.

Problem Setup

$$\max_{x,y} U(x, y)$$

subject to

$$M = p_x x + p_y y,$$

where

- M is income,
- p_x and p_y are prices of goods x and y ,
- x and y are quantities consumed.

Lagrangian Function

We form the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = U(x, y) + \lambda(M - p_x x - p_y y),$$

where λ is the **Lagrange multiplier**.

First-Order Conditions (FOC)

Differentiating the Lagrangian and setting each derivative equal to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = U_x(x, y) - \lambda p_x = 0 \quad (1)$$

$$\frac{\partial \mathcal{L}}{\partial y} = U_y(x, y) - \lambda p_y = 0 \quad (2)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - p_x x - p_y y = 0 \quad (3)$$

Equations (1)–(3) jointly determine the optimal values (x^*, y^*, λ^*) .

Marginal Rate of Substitution and Price Ratio

From (1) and (2), we can eliminate λ :

$$\lambda = \frac{U_x}{p_x} = \frac{U_y}{p_y}$$

which implies

$$\frac{U_x}{U_y} = \frac{p_x}{p_y}.$$

The left-hand side is the **marginal rate of substitution** (MRS) between x and y :

$$MRS_{xy} = \frac{U_x}{U_y}.$$

Thus the utility-maximizing condition can be written as

$$MRS_{xy} = \frac{p_x}{p_y}.$$

Economic meaning: At the optimum, the rate at which the consumer is willing to trade x for y (MRS) equals the rate at which the market allows them to trade x for y (price ratio).

Example: Cobb–Douglas Utility

Let

$$U(x, y) = x^{1/2}y^{1/2},$$

subject to

$$M = p_x x + p_y y.$$

Step 1: Lagrangian

$$\mathcal{L} = x^{1/2}y^{1/2} + \lambda(M - p_x x - p_y y).$$

Step 2: First-order conditions

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/2} - \lambda p_x = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2}x^{1/2}y^{-1/2} - \lambda p_y = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = M - p_x x - p_y y = 0.$$

Divide the first FOC by the second:

$$\frac{\frac{1}{2}x^{-1/2}y^{1/2}}{\frac{1}{2}x^{1/2}y^{-1/2}} = \frac{\lambda p_x}{\lambda p_y} \Rightarrow \frac{y}{x} = \frac{p_x}{p_y}.$$

So

$$y = \frac{p_x}{p_y}x.$$

Substitute into the budget constraint:

$$M = p_x x + p_y \left(\frac{p_x}{p_y} x \right) = 2p_x x \quad \Rightarrow \quad x^* = \frac{M}{2p_x}, \quad y^* = \frac{M}{2p_y}.$$

Thus the optimal bundle splits income equally between the two goods in this symmetric Cobb–Douglas case.

13 Consumer's Equilibrium

Consumer's equilibrium refers to a situation where a consumer allocates income in such a way that utility is maximized subject to a budget constraint. Utility depends on two goods:

$$U = U(x, y)$$

and income constraint:

$$M = p_x x + p_y y.$$

13.1 Deriving Equilibrium Using Total Differentiation

Let the equilibrium utility level be:

$$\bar{U} = U(x, y).$$

Taking total derivative:

$$d\bar{U} = U_x dx + U_y dy.$$

At equilibrium, utility is maximized, so no further increase is possible:

$$d\bar{U} = 0.$$

Thus:

$$U_x dx + U_y dy = 0.$$

Rearranging:

$$U_x dx = -U_y dy.$$

Dividing by dx :

$$\frac{dy}{dx} = -\frac{U_x}{U_y}.$$

The slope of an indifference curve is:

$$MRS_{xy} = \frac{U_x}{U_y}.$$

Hence:

$$-\frac{dy}{dx} = MRS_{xy}.$$

13.2 Total Differentiation of Budget Line

Budget constraint:

$$M = p_x x + p_y y.$$

Income is fixed, so $dM = 0$:

$$0 = p_x dx + p_y dy.$$

Rearranging:

$$p_x dx = -p_y dy.$$

Dividing by dx :

$$\frac{dy}{dx} = -\frac{p_x}{p_y}.$$

This is the slope of the budget line.

13.3 Consumer Equilibrium Condition

At equilibrium:

slope of IC = slope of budget line.

Thus:

$$-\frac{U_x}{U_y} = -\frac{p_x}{p_y}.$$

Cancelling negative signs:

$$\frac{U_x}{U_y} = \frac{p_x}{p_y}.$$

In marginal utility terms:

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y}.$$

Rearranging:

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}.$$

Meaning: *Utility per rupee is equal across goods at equilibrium.*

13.4 Equilibrium with Many Goods

For goods $1, 2, \dots, n$:

$$\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \dots = \frac{MU_n}{p_n}.$$

The last rupee spent on every good yields the same marginal utility.

13.5 Graphical Representation: IC and Budget Line

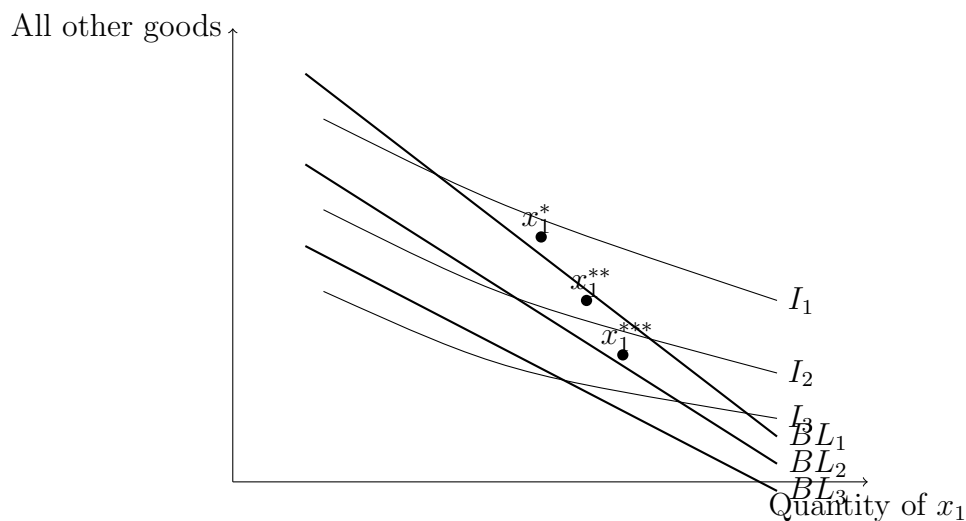


Figure 20: Consumer Equilibrium and Effect of Price Change

Interpretation: When p_1 decreases, the budget line becomes flatter. The consumer moves to a higher indifference curve and consumes more of good x_1 .

14 Supply Curve: Theory

The supply curve is derived from the producer's goal of profit maximization.

Profit:

$$\pi = p \cdot q - C(q),$$

where

- p = market price of the good,
- q = output quantity,
- $C(q)$ = cost function.

A firm increases production as long as:

$$p > MC(q).$$

At optimum:

$$p = MC(q).$$

Since marginal cost typically increases with output, the supply curve slopes upward.

15 Perfect Competition

A perfectly competitive market satisfies:

1. **Large number of buyers:** no single buyer affects price.
2. **Large number of sellers:** no single seller affects price.
3. **Homogeneous product:** all firms sell identical goods.
4. **Free entry and exit:** firms can freely enter or exit the market.

These imply:

Price remains constant for every individual buyer and seller.

Thus buyers and sellers are **price takers**.

16 Worked Numerical Examples

16.1 Example 1: Consumer Equilibrium Using MU per Rupee

A consumer has income $M = 120$ and faces prices

$$p_x = 10, \quad p_y = 20.$$

He can buy two goods x and y . The marginal utilities (MU) from different quantities are:

Quantity	MU_x	MU_x/p_x	Quantity	MU_y	MU_y/p_y
1	50	5.0	1	80	4.0
2	40	4.0	2	60	3.0
3	30	3.0	3	40	2.0
4	20	2.0	4	20	1.0

Table 8: Marginal utility and MU per rupee

The consumer should allocate income so that

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}.$$

From the table, $\frac{MU_x}{p_x} = 4$ when $x = 2$, and $\frac{MU_y}{p_y} = 4$ when $y = 1$. Cost of this bundle:

$$10 \cdot 2 + 20 \cdot 1 = 40 \leq 120.$$

The remaining income can be spent in such a way that MU per rupee remains equal; this demonstrates the equal-marginal principle in practice.

16.2 Example 2: Utility Maximization with Lagrangian

Let

$$U(x, y) = x^{1/2}y^{1/2}, \quad M = 100, \quad p_x = 10, \quad p_y = 20.$$

Step 1: Lagrangian

$$\mathcal{L} = x^{1/2}y^{1/2} + \lambda(100 - 10x - 20y).$$

Step 2: FOC

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{2}x^{-1/2}y^{1/2} - 10\lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2}x^{1/2}y^{-1/2} - 20\lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 100 - 10x - 20y = 0.$$

From the first two equations:

$$\frac{\frac{1}{2}x^{-1/2}y^{1/2}}{\frac{1}{2}x^{1/2}y^{-1/2}} = \frac{10\lambda}{20\lambda} \quad \Rightarrow \quad \frac{y}{x} = \frac{1}{2} \quad \Rightarrow \quad y = \frac{x}{2}.$$

Substitute into budget:

$$100 = 10x + 20\left(\frac{x}{2}\right) = 10x + 10x = 20x \Rightarrow x^* = 5, \quad y^* = \frac{5}{2} = 2.5.$$

So the optimal bundle is $(x^*, y^*) = (5, 2.5)$.

16.3 Example 3: Firm Supply from Profit Maximization

Profit:

$$\pi(q) = pq - C(q).$$

Let

$$p = 50, \quad C(q) = 10q + q^2.$$

Then

$$\pi(q) = 50q - 10q - q^2 = 40q - q^2.$$

FOC:

$$\frac{d\pi}{dq} = 40 - 2q = 0 \Rightarrow q^* = 20.$$

SOC:

$$\frac{d^2\pi}{dq^2} = -2 < 0 \Rightarrow q^* = 20 \text{ yields maximum profit.}$$

So the firm's optimal supply at price 50 is $q^* = 20$ units.

17 Lagrangian Diagram for Consumer Choice

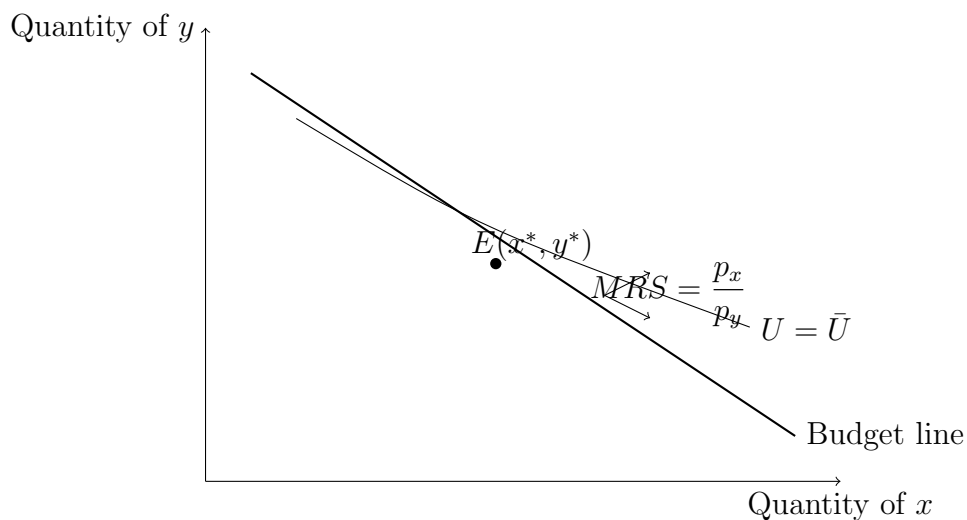


Figure 21: Lagrangian Solution: Tangency of Indifference Curve and Budget Line

18 Supply Curve with MC and ATC

18.1 Firm's Short-Run Supply

In perfect competition, a firm takes price p as given and chooses q such that

$$p = MC(q),$$

as long as price covers average variable cost. The upward-sloping part of the MC curve above the minimum of AVC becomes the firm's supply curve.

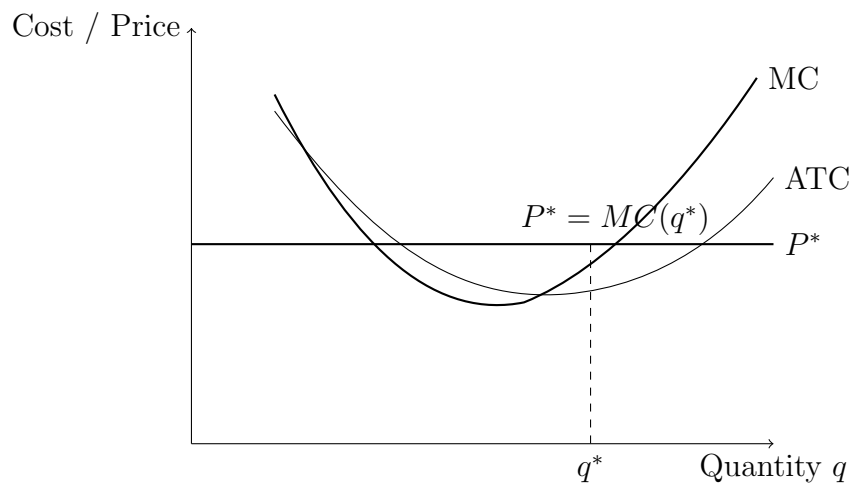


Figure 22: Firm Supply: Intersection of Price with MC above ATC/AVC

19 Consumer Equilibrium: Summary Table

Concept	Condition / Formula
Utility function	$U = U(x, y)$
Budget line	$M = p_x x + p_y y$
Slope of IC	$-\frac{dy}{dx} = -\frac{U_x}{U_y}$
Slope of budget line	$-\frac{dy}{dx} = -\frac{p_x}{p_y}$
Equilibrium condition	$\frac{U_x}{U_y} = \frac{p_x}{p_y}$
MU form	$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}$
Many goods	$\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = \dots = \frac{MU_n}{p_n}$

Table 9: Key Consumer Equilibrium Relationships

20 Revision Notes (Quick Summary)

Demand and Supply

- Law of demand: $P \uparrow \Rightarrow Q_d \downarrow$ (ceteris paribus).
- Law of supply: $P \uparrow \Rightarrow Q_s \uparrow$.
- Equilibrium: $Q_d = Q_s$ at price P^* and quantity Q^* .

Elasticity

- Price elasticity of demand:

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P} = \frac{dQ_d}{dP} \cdot \frac{P}{Q_d}.$$

- Elastic ($|E_d| > 1$), inelastic ($|E_d| < 1$), unitary ($|E_d| = 1$).

Utility and Consumer Equilibrium

- Marginal utility:

$$MU = \frac{\Delta TU}{\Delta Q}, \quad MU \downarrow \text{ as } Q \uparrow.$$

- Consumer equilibrium (two goods):

$$\frac{MU_x}{p_x} = \frac{MU_y}{p_y}.$$

- Lagrangian condition:

$$\frac{U_x}{U_y} = \frac{p_x}{p_y}.$$

Market Equilibrium and Surplus

- Equilibrium price: if $Q_d = a - bP$, $Q_s = c + dP$,

$$P^* = \frac{a - c}{b + d}.$$

- Consumer surplus and producer surplus are triangles above and below P^* .

Production and Supply

- Profit:

$$\pi(q) = pq - C(q).$$

- Profit-maximizing condition:

$$p = MC(q).$$

- In perfect competition, the firm is a price taker and its supply is the upward-sloping part of MC above AVC.

21 Production Theory

Production theory studies how firms convert inputs into output. The general production function is written as:

$$q = f(X_1, X_2, \dots, X_n),$$

where $X_1, X_2, \dots, X_n$ are inputs or factors of production.

21.1 Two-Input Production Function

In microeconomics, we often consider only two inputs:

- Capital (K): machines, equipment, tools, buildings.
- Labour (L): human effort (physical and mental).

Thus, the production function becomes:

$$q = F(K, L),$$

which represents the maximum output attainable from given inputs under fixed technology.

21.2 Assumptions of the Production Function

- Technology is constant.
- Inputs are substitutable to some degree.
- The firm seeks to produce at maximum efficiency.
- Inputs exhibit diminishing marginal productivity.

21.3 Isoquant Diagram

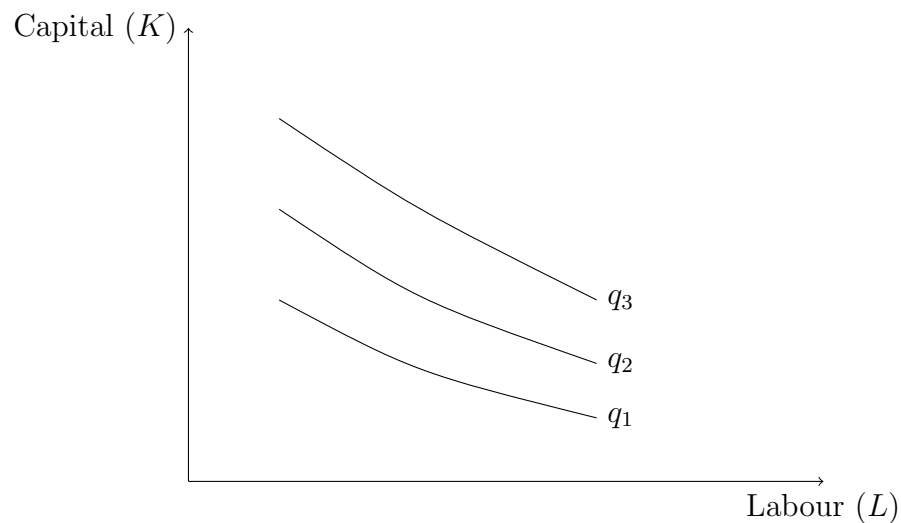


Figure 23: Isoquants showing different levels of output. Higher isoquants represent higher output.

Isoquants show all combinations of capital and labour that yield the same output.

21.4 Examples of Production Functions

1. Linear (Perfect Substitutes):

$$q = 4K + 2L.$$

2. Leontief (Fixed Proportion):

$$q = \min(2K, 4L).$$

3. Cobb–Douglas:

$$q = K^\alpha L^{1-\alpha}.$$

Each functional form represents a different technological relationship between inputs and outputs.

22 Isoquants and MRTS

Isoquants represent all combinations of capital (K) and labour (L) that produce the same level of output:

$$q = F(K, L).$$

22.1 Properties of Isoquants

- **Downward sloping:** If labour increases, capital must decrease to keep output constant.
- **Convex to the origin:** MRTS diminishes along the isoquant, making it convex.
- **Higher isoquants imply higher output:** $q_3 > q_2 > q_1$.
- **Isoquants do not intersect:** A single input combination cannot yield two different output levels.

22.2 Isoquant Diagram

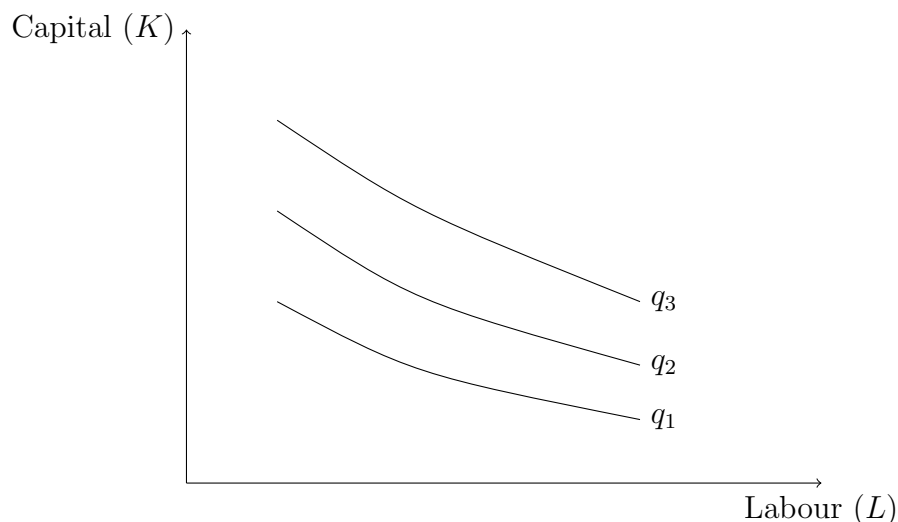


Figure 24: Isoquants showing different levels of output

22.3 Marginal Rate of Technical Substitution (MRTS)

Along an isoquant, output is constant:

$$dq = F_K dK + F_L dL = 0.$$

Rearranging:

$$-\frac{dK}{dL} = \frac{F_L}{F_K} = \frac{MP_L}{MP_K}.$$

Thus, MRTS is defined as:

$$MRTS_{KL} = \frac{MP_L}{MP_K}.$$

It measures how much capital must be reduced when labour is increased by one unit to maintain the same output.

22.4 Law of Diminishing MRTS

As labour increases and capital decreases along an isoquant:

$$MRTS_{KL} \downarrow.$$

This occurs due to diminishing marginal productivity of labour and increasing marginal productivity of capital.

22.5 Example

Given production function:

$$q = K^{1/2}L^{1/2},$$

we have:

$$MP_K = \frac{1}{2}K^{-1/2}L^{1/2}, \quad MP_L = \frac{1}{2}K^{1/2}L^{-1/2}.$$

Thus:

$$MRTS_{KL} = \frac{MP_L}{MP_K} = \frac{K}{L}.$$

If $K = 10$ and $L = 5$, then:

$$MRTS_{KL} = 2.$$

23 Cost Minimisation and $MRTS = w/r$

A firm aims to produce a given level of output at the lowest possible cost. Let K denote capital, L denote labour, and let r and w be their respective prices. Total cost is:

$$C = rK + wL.$$

The firm chooses K and L to minimise cost subject to the production requirement:

$$q_0 = F(K, L).$$

23.1 Isocost Line

The isocost line shows all input combinations that cost the same amount:

$$C = rK + wL.$$

Solving for K :

$$K = \frac{C}{r} - \frac{w}{r}L.$$

Its slope is:

$$\text{slope} = -\frac{w}{r}.$$

23.2 Condition for Cost Minimisation

At the least-cost combination, the isoquant is tangent to the isocost line:

$$MRTS_{KL} = \frac{w}{r}.$$

Since:

$$MRTS_{KL} = \frac{MP_L}{MP_K},$$

we obtain the tangency condition:

$$\frac{MP_L}{MP_K} = \frac{w}{r} \quad \text{or} \quad \frac{MP_L}{w} = \frac{MP_K}{r}.$$

This implies that the marginal product per rupee spent on each input must be equal.

23.3 Cost Minimisation Diagram

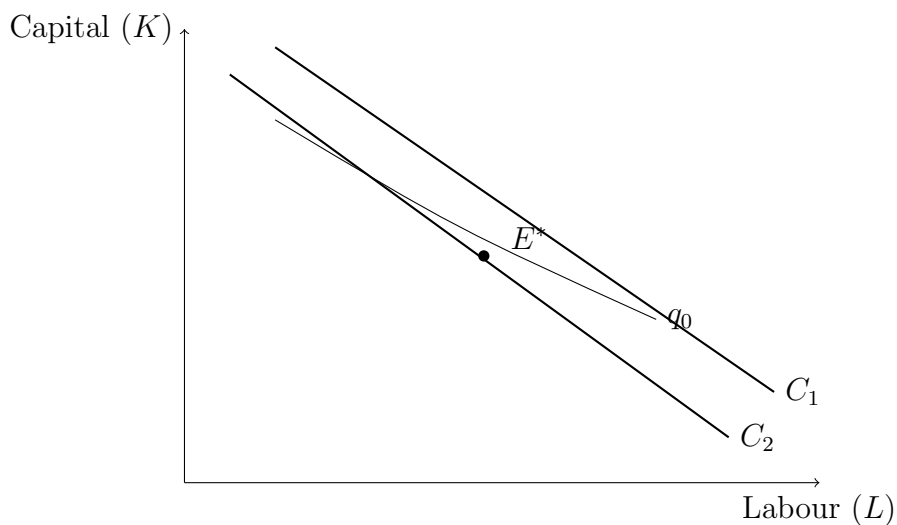


Figure 25: Cost minimisation at tangency of isoquant and isocost line

23.4 Numerical Example

Given:

$$q = K^{1/2}L^{1/2}, \quad q_0 = 20, \quad w = 10, \quad r = 20.$$

MRTS:

$$MRTS_{KL} = \frac{K}{L}.$$

Cost-minimising condition:

$$\frac{K}{L} = \frac{w}{r} = \frac{1}{2} \Rightarrow K = \frac{L}{2}.$$

Substituting into production function:

$$20 = \left(\frac{L}{2}\right)^{1/2} L^{1/2} = \frac{L}{\sqrt{2}} \Rightarrow L = 20\sqrt{2}.$$

Thus:

$$L^* \approx 28.28, \quad K^* \approx 14.14.$$

24 Cobb–Douglas Production Function and Returns to Scale

The Cobb–Douglas production function is one of the most widely used functional forms in economics:

$$Q = AK^\alpha L^\beta,$$

where A denotes technology, K is capital, L is labour, and α, β are output elasticities.

24.1 Interpretation of Parameters

- A : Technology parameter. Higher A implies higher output for the same inputs.
- α : Output elasticity of capital.
- β : Output elasticity of labour.

Taking logs:

$$\ln Q = \ln A + \alpha \ln K + \beta \ln L,$$

which is linear in logs and widely used in empirical estimation.

24.2 Returns to Scale

Scale all inputs by a factor t :

$$Q' = A(tK)^\alpha (tL)^\beta = t^{\alpha+\beta} Q.$$

Thus:

$$\begin{cases} \alpha + \beta = 1 & \text{Constant returns to scale (CRS),} \\ \alpha + \beta > 1 & \text{Increasing returns to scale (IRS),} \\ \alpha + \beta < 1 & \text{Decreasing returns to scale (DRS).} \end{cases}$$

Example:

$$Q = K^{0.4} L^{0.6} \quad (\alpha + \beta = 1) \Rightarrow \text{CRS.}$$

24.3 Technology Shifts and Isoquants

An increase in the technology parameter shifts the isoquant inward:

$$A_1 < A_2 < A_3.$$

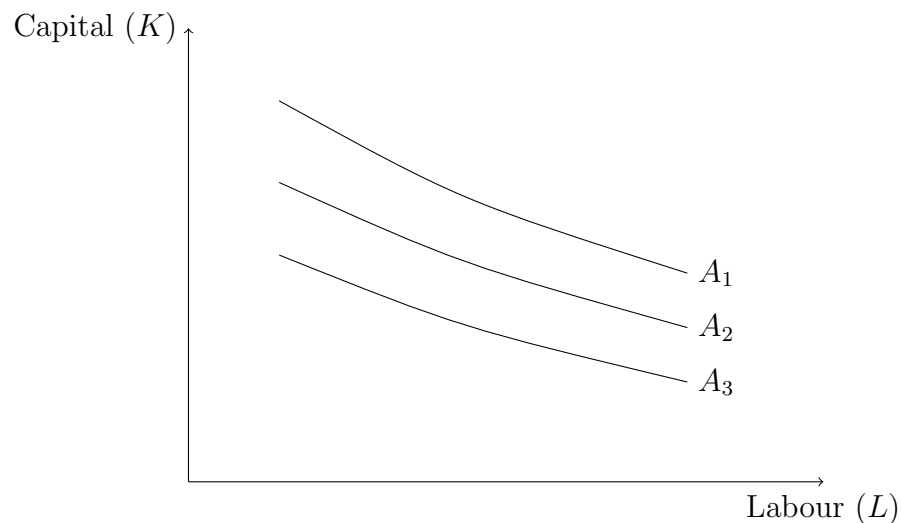


Figure 26: Higher technology shifts the isoquant inward

24.4 Example: Effect of Technology

Let $Q = AK^{0.5}L^{0.5}$ and $K = 4$, $L = 9$.

$$Q = A(2)(3) = 6A.$$

If $A = 10$:

$$Q = 60.$$

If $A = 15$:

$$Q = 90.$$

Thus, output increases solely due to higher technology.

25 Perfect Substitutes and Perfect Complements

25.1 Perfect Substitutes

Two inputs K (capital) and L (labour) are perfect substitutes if one input can completely replace the other at a constant rate. The production function is:

$$Q = aK + bL,$$

where a and b are the contributions of capital and labour.

Isoquants are straight lines because the marginal rate of technical substitution is constant:

$$MRTS_{KL} = \frac{a}{b}.$$

Isoquant for Perfect Substitutes

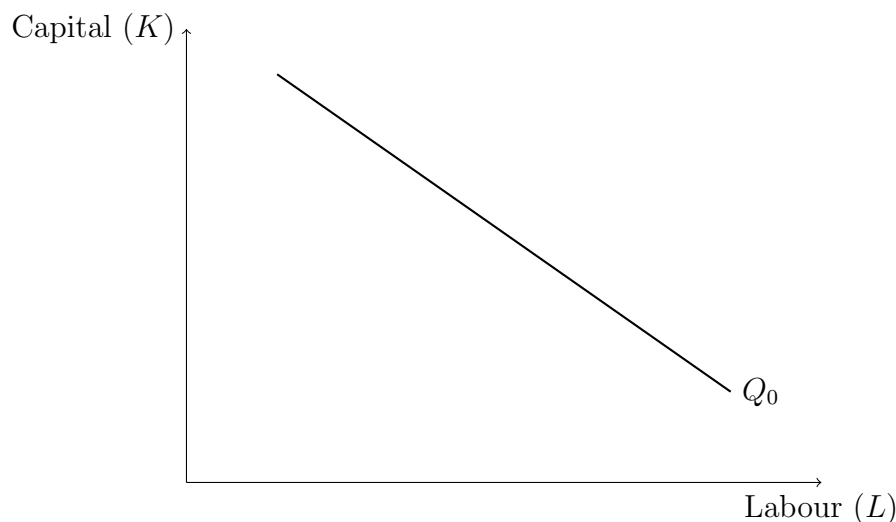


Figure 27: Straight-line isoquant for perfect substitutes

Optimal Choice

The firm chooses the cheaper input:

$$\frac{a}{b} \begin{cases} > \frac{w}{r} & \Rightarrow \text{Use all labour,} \\ < \frac{w}{r} & \Rightarrow \text{Use all capital,} \\ = \frac{w}{r} & \Rightarrow \text{Any combination of inputs.} \end{cases}$$

Example: If $Q = 2K + L$ and labour is cheaper, the firm uses only labour.

25.2 Perfect Complements (Leontief)

Inputs are perfect complements when they must be used in fixed proportions:

$$Q = \min(aK, bL).$$

Extra units of one input cannot substitute for the other.

Isoquant for Perfect Complements

Isoquants are L-shaped because substitution is impossible.

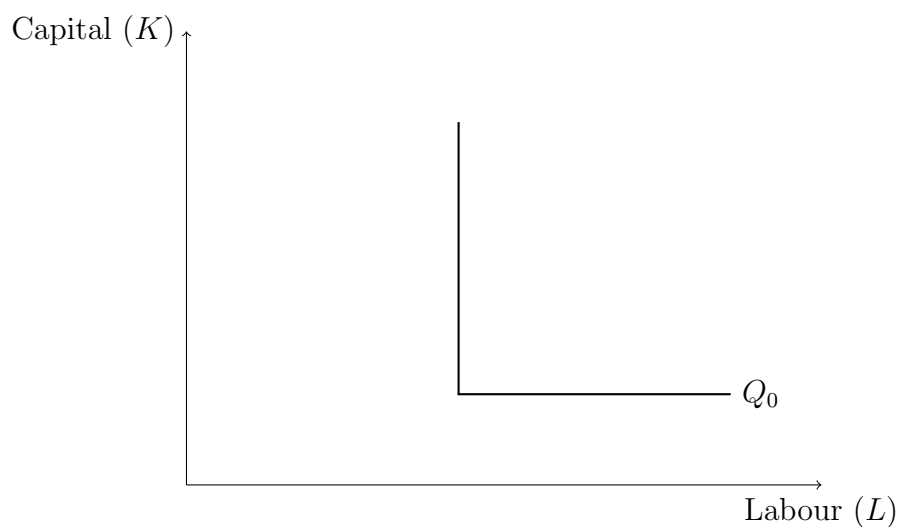


Figure 28: L-shaped isoquant for perfect complements

Example

If:

$$Q = \min(K, 2L),$$

to produce $Q = 10$, the firm requires:

$$K = 10, \quad L = 5.$$

Any extra K or L beyond this ratio does not increase output.

26 Short-Run and Long-Run Costs

26.1 Short-Run Costs

In the short run, at least one input is fixed. Total cost is the sum of fixed and variable cost:

$$TC = TFC + TVC.$$

- **Total Fixed Cost (TFC)** is constant.
- **Total Variable Cost (TVC)** increases with output.
- **Total Cost (TC)** shifts TVC upward by TFC.

26.2 Average and Marginal Costs

$$AFC = \frac{TFC}{q}, \quad AVC = \frac{TVC}{q}, \quad AC = \frac{TC}{q} = AFC + AVC.$$

Marginal cost is:

$$MC = \frac{dTC}{dq} = \frac{dTVC}{dq}.$$

MC intersects AVC and AC at their minimum points.

26.3 Short-Run Cost Curve Diagram

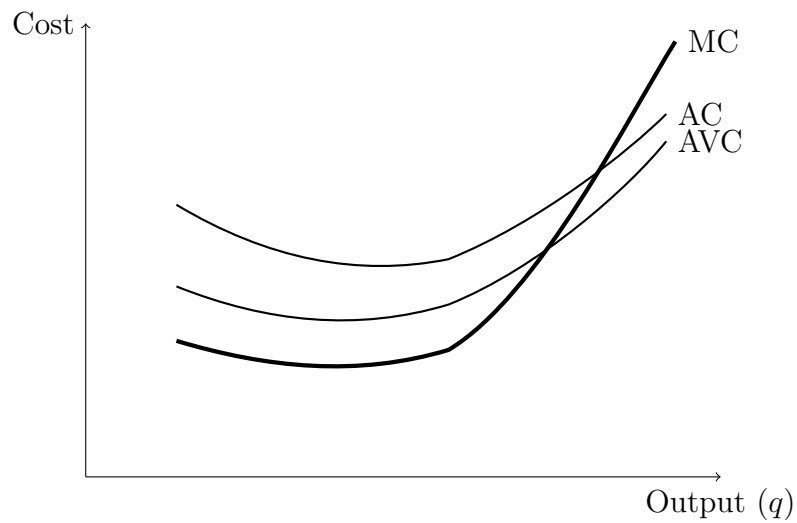


Figure 29: Short-run MC, AVC, and AC curves

26.4 TFC, TVC, and TC Curves

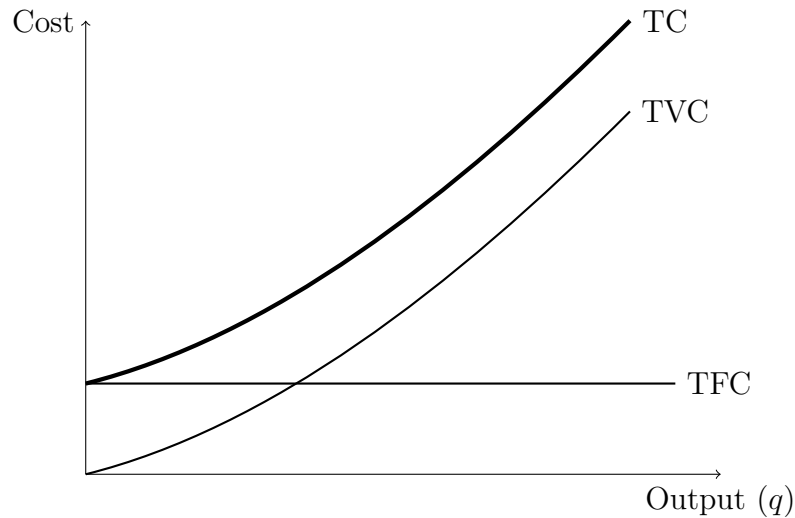


Figure 30: TFC, TVC, and TC curves

26.5 Long-Run Costs

In the long run, all inputs are variable. The long-run average cost (LRAC) curve represents the minimum cost of producing each output level:

$$LRAC = \min_{K,L} \frac{C}{q}.$$

LRAC is the **envelope** of short-run average cost curves.

26.6 LRAC and SRAC Diagram

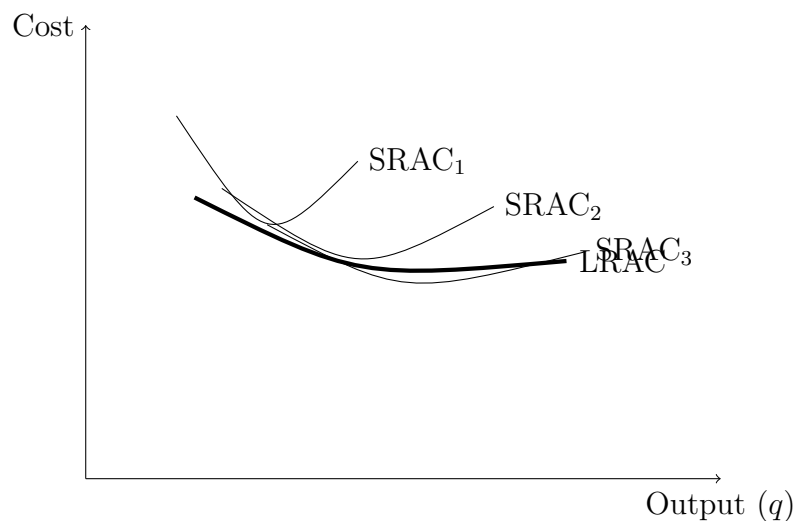


Figure 31: LRAC as the envelope of SRAC curves

26.7 Economies and Diseconomies of Scale

- **Economies of scale:** LRAC decreases as output increases.
- **Constant returns:** LRAC is flat.
- **Diseconomies of scale:** LRAC increases as output increases.

27 Perfect Competition

A perfectly competitive market has the following features:

- Large number of buyers and sellers.
- Homogeneous product.
- Free entry and exit.
- Perfect information.
- Firms are price takers.

27.1 Price Taking and Revenue

Since the firm is a price taker:

$$P = MR = AR.$$

The firm's demand curve is perfectly elastic at the market price.

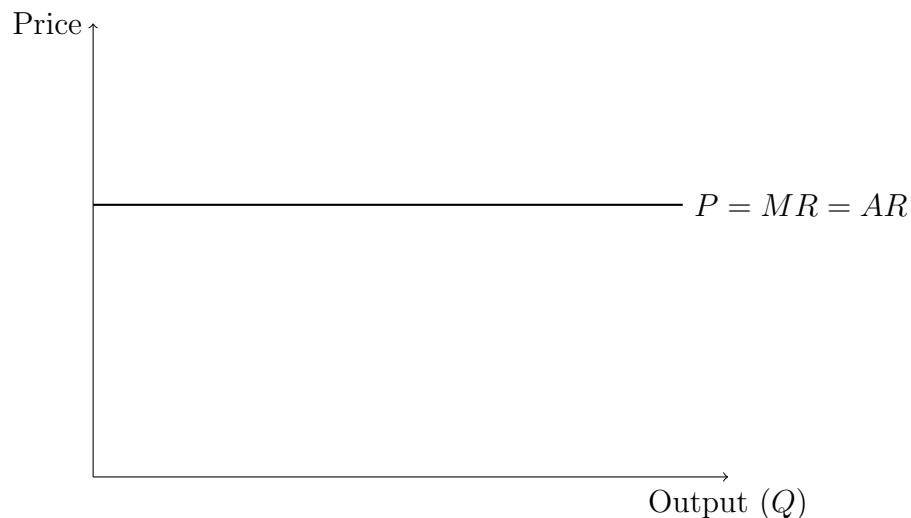


Figure 32: Perfectly elastic demand for the firm

27.2 Profit Maximisation: $P = MC$

The firm chooses output Q to maximise:

$$\pi = PQ - C(Q).$$

First-order condition:

$$MR = MC.$$

Since $MR = P$:

$$P = MC.$$

27.3 Short-Run Equilibrium of the Firm

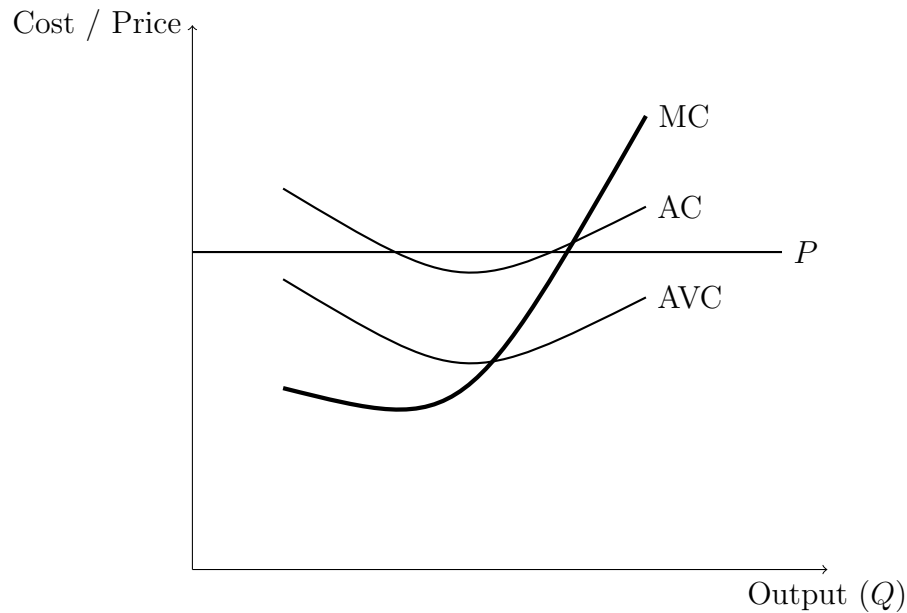


Figure 33: Short-run equilibrium: $P = MC$

27.4 Shutdown Condition

The firm shuts down if:

$$P < AVC_{\min}.$$

27.5 Firm's Supply Curve

The short-run supply curve is the portion of the marginal cost curve above the minimum of AVC.

27.6 Long-Run Equilibrium

With free entry and exit:

$$P = MC = AC_{\min}.$$

Firms earn zero economic profit in the long run.

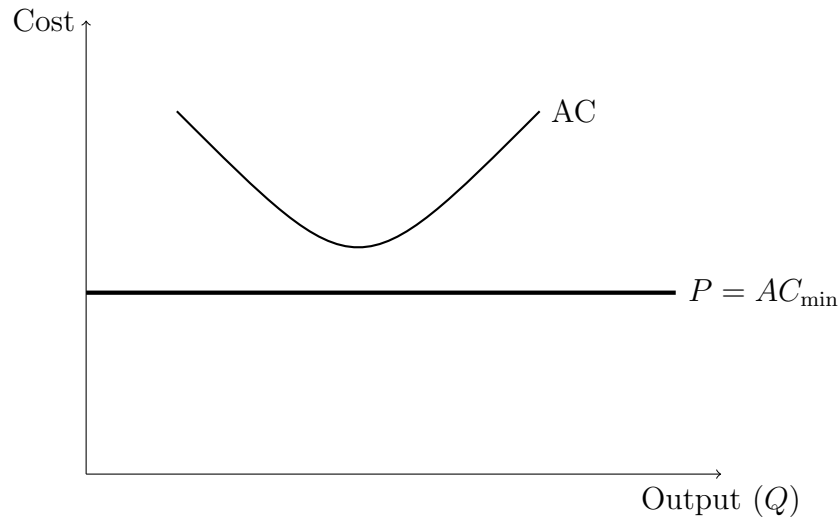


Figure 34: Long-run equilibrium in perfect competition

28 Profit Maximisation and Shutdown Decision

In a perfectly competitive market, the firm is a price taker. Therefore, market price p is constant and:

$$P = MR = AR.$$

The firm chooses output Q to maximise profit.

28.1 Profit Function

Profit is defined as:

$$\pi(Q) = pQ - C(Q),$$

where:

- p is the market price of the good,
- Q is the output,
- $C(Q)$ is the total cost of producing Q units.

The firm's objective is:

$$\max_Q \pi(Q).$$

28.2 First-Order Condition (FOC)

To maximise profit, take the first derivative and set it equal to zero:

$$\frac{d\pi(Q)}{dQ} = 0.$$

Differentiate the profit function:

$$\frac{d}{dQ}(pQ - C(Q)) = p - C'(Q).$$

Since $C'(Q) = MC(Q)$ (marginal cost), we get:

$$p - MC(Q) = 0.$$

Thus, the profit-maximising condition is:

$$\boxed{P = MC(Q^*)}.$$

This means the firm produces the quantity at which price equals marginal cost.

28.3 Second-Order Condition (SOC)

The first-order condition gives a critical point. To ensure that it maximises profit, we need:

$$\frac{d^2\pi}{dQ^2} < 0.$$

Since:

$$\frac{d^2\pi}{dQ^2} = -MC'(Q),$$

the SOC requires:

$$MC'(Q) > 0.$$

Thus, the marginal cost curve must be rising at the optimal level of output.

28.4 Profit Maximisation Diagram

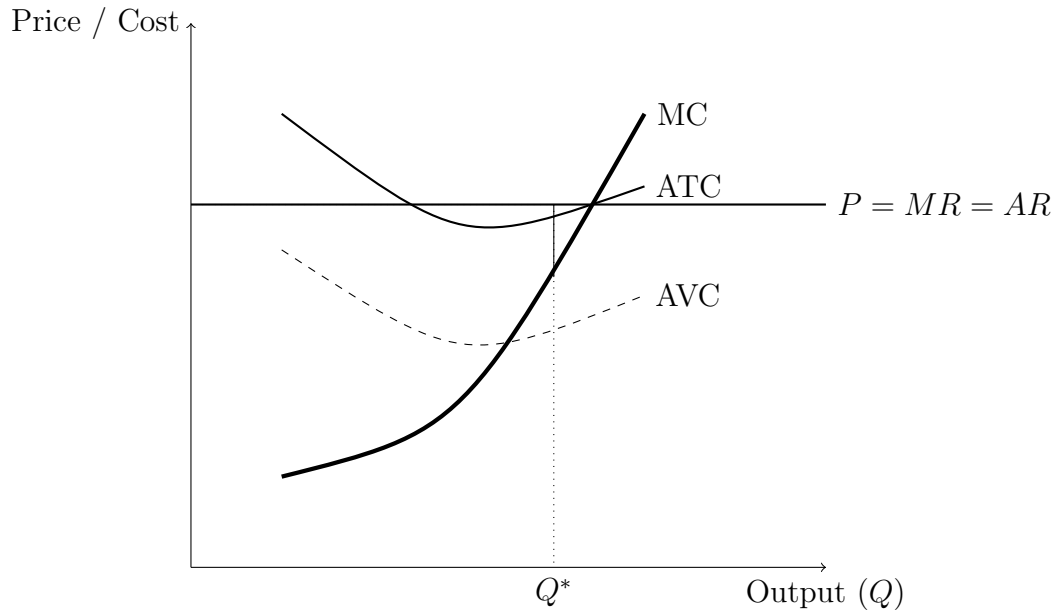


Figure 35: Profit maximisation under perfect competition: $P = MC$

28.5 Total Revenue, Total Cost, and Profit

Total revenue:

$$TR = pQ.$$

Total cost:

$$TC = ATC \cdot Q.$$

Profit:

$$\pi = TR - TC = (p - ATC)Q.$$

Graphically:

- Area under price line = total revenue.
- Area under ATC curve = total cost.
- The rectangle between price and ATC at Q^* = profit.

28.6 Shutdown Decision

A firm decides whether to produce or temporarily shut down in the **short run** based on the relationship between price and average variable cost.

Continue Producing

The firm should continue to operate if:

$$P \geq AVC.$$

Even if the firm makes a loss (if $ATC > P$), it continues producing as long as it covers variable costs.

Shutdown

The firm shuts down if:

$$P < AVC.$$

In this case, the firm cannot cover its variable costs, so producing would increase losses.

28.7 Summary of Cases

1. **Profit:** $P > ATC$ Firm earns positive profit.
2. **Loss but operate:** $AVC < P < ATC$ Firm covers variable cost and part of fixed cost; continues.
3. **Shutdown:** $P < AVC$ Firm produces zero output.

29 Industry–Firm Equilibrium, Profit, Loss and Long-Run Adjustment

In perfect competition, the industry determines the market price, and the firm takes that price as given. The industry price comes from the intersection of market demand and market supply. Given this price, the individual firm chooses its profit-maximising output where marginal revenue equals marginal cost.

29.1 Step 1: Industry Equilibrium and Firm Price

For the industry, the equilibrium price P^0 and quantity Q^0 are determined by:

$$S(Q) = D(Q).$$

At this point, supply equals demand.

An individual competitive firm is a price taker, so it faces a horizontal demand curve at the market price:

$$P^0 = MR = AR.$$

The firm chooses output Q^* where:

$$MR = MC \quad \Rightarrow \quad P^0 = MC(Q^*).$$

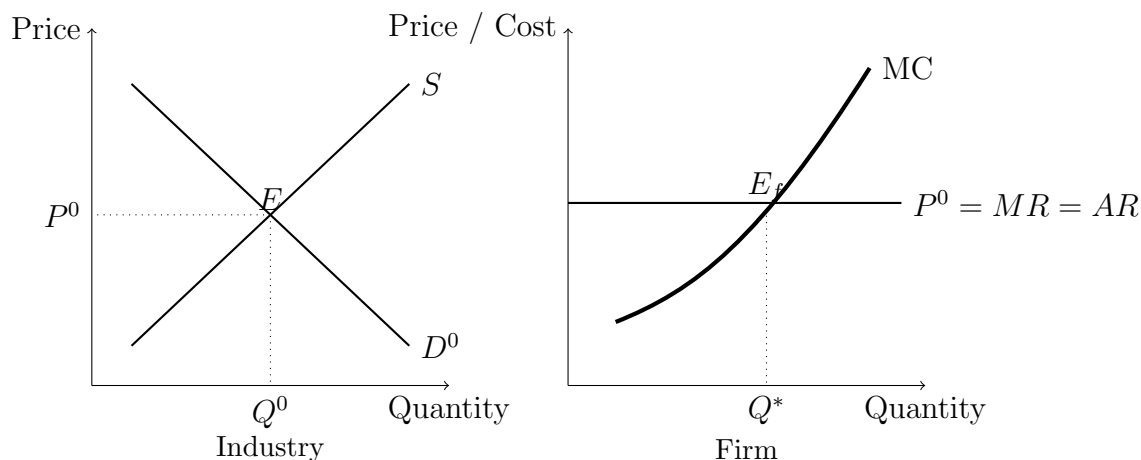


Figure 36: Industry equilibrium and the competitive firm's price-taking behaviour

29.2 Step 2: Adding ATC and AVC to Determine Profit or Loss

To determine whether the firm earns profit or loss, we compare the price with average total cost (ATC) and average variable cost (AVC) at Q^* .

- If $P^0 > ATC(Q^*)$, the firm earns positive profit.
- If $AVC(Q^*) < P^0 < ATC(Q^*)$, the firm incurs a loss but continues to operate.
- If $P^0 < AVC(Q^*)$, the firm shuts down in the short run.

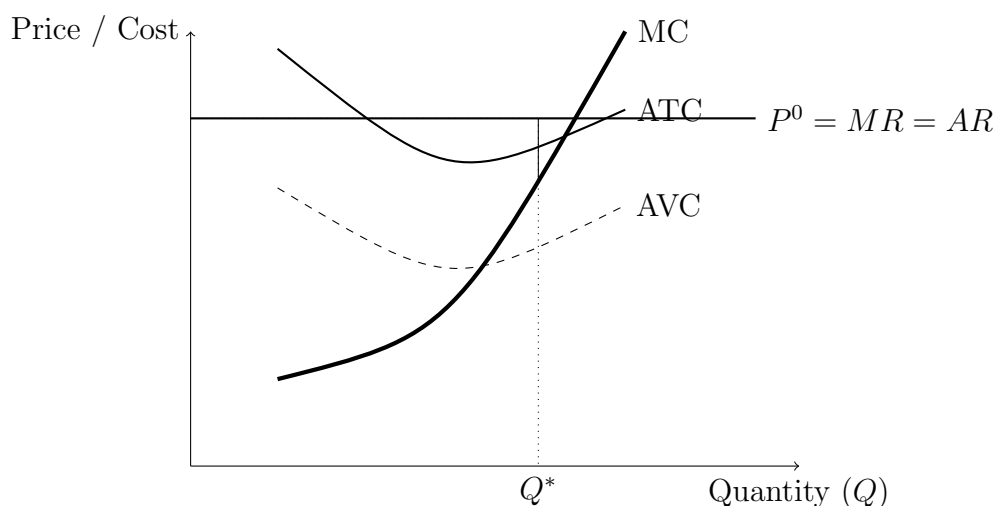


Figure 37: Firm equilibrium with ATC and AVC: profit if $P^0 > ATC$

The shaded area between the price line and the ATC curve represents profit per unit times quantity.

29.3 Step 4: Total Revenue, Total Cost and Profit

At the equilibrium output Q^* and price P^0 :

$$TR = P^0 Q^*, \quad TC = ATC(Q^*) Q^*.$$

Profits are:

$$\pi = TR - TC = (P^0 - ATC(Q^*)) Q^*.$$

Graphically:

- The rectangle under the price line up to Q^* is total revenue.
- The rectangle under ATC up to Q^* is total cost.
- The area between them is profit.

29.4 Step 6–9: Demand Decreases and the Firm Makes a Loss

Suppose industry demand shifts left from D^0 to D^1 . The new intersection with supply yields a lower price P^1 and lower output Q^1 for the industry.

The individual firm now faces a lower price line P^1 . It chooses output Q_1 where:

$$P^1 = MC(Q_1).$$

At this new quantity, if:

$$AVC(Q_1) < P^1 < ATC(Q_1),$$

the firm incurs a loss but continues to produce, because it can still cover its variable costs.

29.5 Shutdown Rule (Extreme Negative Case)

If demand falls further and the price drops below average variable cost, i.e.

$$P^2 < AVC(Q),$$

for all relevant levels of Q , then the firm is better off shutting down in the short run.

- If $P < AVC_{\min}$, the firm shuts down (produces $Q = 0$).
- If $AVC < P < ATC$, the firm produces but incurs a loss.

29.6 Long-Run Equilibrium of a Competitive Industry

In the long run:

- All inputs are variable; there is no fixed cost.
- Firms can freely enter or exit the industry.
- If firms are making economic profits, new firms enter, increasing supply and driving price down.
- If firms are making losses, some firms exit, reducing supply and driving price up.

Long-run competitive equilibrium occurs when:

$$P = LMC = LAC = AR = MR,$$

and firms earn zero economic profit (normal profit).

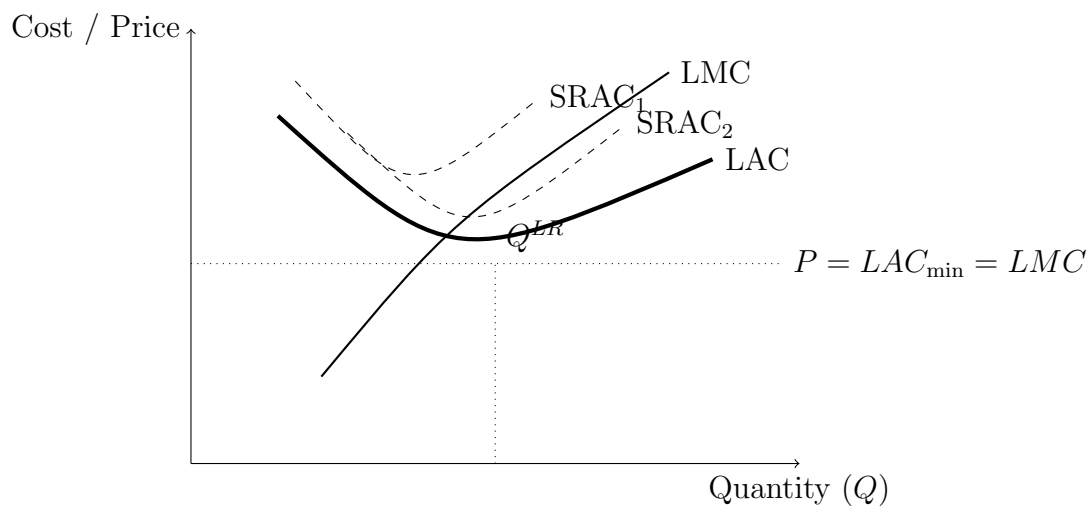


Figure 38: Long-run equilibrium in perfect competition: $P = LMC = LAC$

In this situation, the firm is operating at the most efficient scale, with zero economic profit but full recovery of all costs including normal return on capital.